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Project Number:	13
Project Title:	Aged Care Compliance Document Summariser
Workshop Time:	02:00 – 05:00 PM
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Main Functional Requirements (Minimum 5)	
1.	Document Upload and Processing (PDF and DOCX – AI/NLP analyse the documents)
2.	AI-Based Document Summarisation (summarize document with NLP and LLM)
3.	Compliance Obligation Extraction (identifies the things that people or the organisation are required to do)
4.	Risk Identification (identify risks – high, medium, low)
5.	Deadline and Responsibility Extraction (who needs to perform an action and when it needs to be completed)
6.	Action Item Identification (extract required actions with Where possible, each action will include the responsible role, timeframe, priority, and source section)
7.	Compliance Summary Dashboard
8.	Chatbot (if needed)

Main Non-functional Requirements and how these are going to be addressed	
Non-functional Requirement	How to address?
Accuracy	The AI-generated summaries and extracted compliance information will be evaluated against manually created data. Precision, recall, F1-score and human evaluation will be used where appropriate.
Usability	A simple web-based dashboard will be developed with clear sections for summaries, obligations, risks, deadlines, responsibilities and action items. The interface will minimise unnecessary complexity for aged-care staff and managers.
Performance	The system will use efficient document-processing and NLP techniques to process documents within a reasonable response time. Large documents will be divided into manageable sections before AI processing.
Security	The prototype will use public or synthetic documents rather than confidential resident information. Uploaded documents and generated results will be handled using appropriate access and storage controls.
Privacy	No personally identifiable resident or staff information will be required for testing. Sensitive information will be removed from any anonymized datasets before processing.
Reliability	The system will include errors handling for unsupported files, missing text, extraction failures and incomplete AI

	responses. Important outputs will include source references for verification.
Responsible Model	The system will clearly state that ML-generated results are decision-support information and must be verified against the original document. The system will not make final legal and compliance decisions.

Resources required (including data) and how to obtain	
Resource	How to obtain
Aged-care compliance documents (Dataset)	Obtain publicly available policies, standards, procedures, guidelines and compliance documents from Australian government and recognised aged-care organisations.
Synthetic compliance documents	Create synthetic aged-care policies and procedures containing predefined obligations, risks, deadlines, responsibilities and action items for controlled testing.
Anonymised documents	Use documents only where personally identifiable or confidential information has been removed and appropriate permission exists.
Programming tool	Use Python, VS Code and relevant development libraries for implementing the AI/NLP pipeline.
Frontend	Use React.js to develop the web-based document upload and compliance summary dashboard.
Backend	Use Python and FastAPI to implement document processing, AI services and API communication.
NLP/ AI Libraries	Use appropriate Python libraries such as Hugging Face Transformers, spaCy, Sentence Transformers and other suitable NLP tools.
Document Processing Libraries	Use libraries such as PyMuPDF and python-docx to extract text from PDF and DOCX files. OCR can be investigated for scanned documents.
Database	Use PostgreSQL or another suitable database to store document metadata, extracted information and evaluation results.
LLM Service	Use an appropriate LLM API or locally hosted model, subject to university requirements, cost, privacy and availability.
Evaluate dataset	Manually annotate selected documents to create ground-truth information for obligations, risks, deadlines, responsibilities and action items.
Development and testing environment	Use a local computer, VS Code, Python environment and Docker where required for reproducible development and deployment.

Any Notes/Requests:

The project aims to develop an AI/NLP-based system that summarises aged-care compliance documents and extracts key obligations, risks, deadlines, responsibilities, and actions. The system will process public, synthetic, or anonymised documents and display the results through a simple dashboard. The project will evaluate the accuracy and usefulness of the AI summaries while considering privacy, hallucination, explainability, and human verification.